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Project Title:	Cytoplasmic/Golgi interactions in the nanofabrication of biomineral composites - a genetic approach using coccolithophores		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Comments:			
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Contact Monitor:		**Contact Phone:**	
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	1. The first task is to identify macromolecular elements associated with the cytoplasm and Golgi which regulate the nucleation and growth of coccolith crystals (calcite). Our approach is to generate and isolate coccolithophore mutants using RNAi which do not express a particular macromolecule - one with a putative mineralizing function - and determine how its absence affects the structure of the biocomposite in vivo. 2. The second task is to explore the effects of nucleating agents and crystal shaping elements identified in the first task on calcite nucleation and growth in vitro - both in bulk solution and in synthetic liposomes. 3. The third task is a study of coccolithophore phase transition between calcifying and noncalcifying morphotypes.		

Task Description:	The overall goal of this project is to determine how CaCO3 formation is regulated in coccolithophores, calcifying phytoplankton responsible for the majority of the CaCO3 formed in the open ocean since the early Cretaceous. Coccolithophores impact the ocean carbon cycle and may regulate the Quaternary pCO2 flux through transitions between calcifying and noncalcifying morphotypes. An understanding of the cellular processes, genetic elements, and environmental cues which regulate the expression of calcifying versus noncalcifying phases will be essential for predicting future climate change. Regulation of CaCO3 crystal size and shape by coccolithophores is relevant to nanofabrication, as biological processes can provide inspiration for development of novel and useful materials such as molecular sieves and catalysts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biomineralization in coccolithophores is a phenomenon of global import. The appearance of coccolithophores and foraminifera — calcareous nannoplankton — in the late Triassic marked the onset of calcium carbonate sedimentation in the deep ocean, a process thought to stabilizes the carbon cycle today. Coccolithophores are the major produces of pelagic CaCO3 and impact the atmosphere by generating CO2 during calcification. Ca2+ + 2HCO3–= CaCO3 + CO2 + H2O. The global carbon cycle is a major societal concern because the extraction and burning of fossil fuels is transferring carbon from the slowing exchanging lithosphere into the atmosphere as CO2, a greenhouse. Theoretically global mean temperature will rise as atmospheric pCO2 increases, a theory consistent with ice core analyses showing a tight correlation between atmospheric pCO2 and sea surface temperature over the past 400,000 years. The ocean is the major reservoir for absorbing atmospheric CO2 changes; the gas is partitioned between atmosphere and ocean through planktonic photosynthesis and CaCO3 formation and by ocean circulation with the exchange of carbon between deep and surface waters. An understanding of how coccolithophore calcification impacts the natural Quaternary pCO2 flux and is impacted by anthropogenic CO2 emissions will be essential for prediction of future climate change.
Task Progress:	The goal of this year's effort was to construct vectors for driving the expression of foreign genes in the coccolithophore Pleurochrysis carterae. To identify an effective group of promoter and terminator elements we began by sequencing the fucoxanthin chlorophyll a/c-binding protein genes (fcp) along with their flanking genomic DNA, since a partial Pleurochrysis fcp cDNA sequence was already available in GenBank. The FCPs are chloroplast proteins encoded in the nuclear genome. We found five fcp genes (fcpA, fcpB, fcpC, fcp0, and fcp0i). All are intronless and have homologous coding sequences of 624 nucleotides. When the cells are grown under low continuous illumination (7.5 micromoles photons/m2/s), each fcp gene is expressed at a relatively high level — greater than that of actins type 1, 2, and 4 combined. mRNA levels of the fcp genes A, C, 0 and 0i are similar and about three times more abundant than the fcpB transcript. The genomic arrangement of fcpA, fcpB, and fcpC has not been determined. fcp0 and fcp0i are arranged head-to-head on opposites strands with 284 and 362 base pairs separating their transcription and translation start sites, respectively. This arrangement of gene pairs is termed bidirectional, and the region between the two transcriptional start sites is a putative bidirectional promoter. Because their coding sequences are so similar (99.2% nucleotide identity), fcp0 and fcp0i probably arose through a recent gene duplication by one of the intra- or intermolecular modes of large DNA palindrome formation. Pair wise comparisons of all identified Pleurochrysis fcp genes — other than the fcp0/fcp0i pair — give a nucleotide identity range of 88.6-93.8%. The bidirectional promoter shared by palindromic fcp0/fcp0i is not itself a palindrome but it does contain the sequence AAATATTTAATT in both orientations separated by 88 base pairs. This sequence may be a cis-acting element as similar sequences are also contained in the promoter region of fcpA, fcpB, and fcpC. The consensus sequence of this fcp AT-rich element is AAATAT(A,T)(A,T)ANT(A,T). The AT-rich element may determine the direction of transcription otherwise it is unclear why the fcp0/fcp0i bidirectional promoter should contain this particular sequence in both orientations. This sequence may be a variation of the TATA box element; however, unlike the TATA box which occurs about 25-35 nucleotides up stream of the transcription start site, the fcp promoter element is located 83-357 nucleotides upstream. It appeared that the fcp promoters would be an excellent choice for driving the expression of foreign genes in Pleurochrysis because 1) promoters of homologous diatom fcp genes effectively drive expression of foreign genes in diatoms, 2) the fcp genes are intronless indicating that all elements required for expression may be included in their promoter and terminator regions, and 3) it should be possible to simultaneously express two different transgenes placed head-to-head on opposite strands separated by the bidirectional fcp0/fcp0i promoter. Vectors were constructed containing fcp promoters (including the 5'UTRs) and terminators flanking the zeocine-resistant she ble gene — which we codon optimized for expression in Pleurochrysis — and the green fluorescent protein gene (gfp) that had been optimized for expression in the green alga Chlamydomonas and had a codon usage similar to that of Pleurochrysis genes. Pleurochrysis cells were stripped of their coccospheres to expose the plasma membrane and supercoiled or linearized vectors were introduced by electroporation. After transfection the cells were incubated at low constant illumination to induce high expression of transgenes driven by the fcp promoters. Thus far expression levels of gfp and zeocine-resistance have been disappointingly low. Our thinking now is that additional elements influence some stage(s) in the mRNA metabolism — transcription, processing, nuclear export, transcription — of native fcp genes which are located within the coding sequences. Thus our next transfection experiments will include vector constructs designed for expression of fcp-gfp and fcp-she ble chimeras. If the chimeras can be efficiently expressed, then expression of transgenes alone should be possible from constructs containing shorter sections of the fcp coding region (minus any ATG sequences) incorporated into the fcp promoter region between the transcription and translation start sites
Bibliography Type:	Description: (Last Updated: 09/11/2019)
Books/Book Chapters	Marsh ME. "Regulation of coccolith calcification in Pleurochrysis carterae." in "Handbook of Biomineralization: Biological Aspects and Structure Formation." Ed. E. Baeuerlein. Weinheim : Wiley-VCH, 2007. p. 211-226. https://doi.org/10.1002/9783527619443.ch12 , May-2007